Algebraic Number Theory NMAL431
Exercise Sheet 4

1 December 2026

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1. (*Krasner's lemma*) Let p be a prime, and let K be a finite extension of $\mathbb{Q}_p$. Let E/K be a finite Galois extension with valuation $v_E : E^\times \to \mathbb{Z}$, and let $x, y \in E$. Let $y_1, \ldots, y_n$ be the Galois conjugates of y over K (taking $y_1 := y$), and assume that $v_E(x - y) > v_E(x - y_i)$ for each $i = 2, \ldots, n$. Prove that $K(y) \subset K(x)$.

2. Let p be a prime, and let $\overline{\mathbb{Q}}_p$ be an algebraic closure of $\mathbb{Q}_p$. Recall that if $K/\mathbb{Q}_p$ is a finite extension contained inside $\overline{\mathbb{Q}}_p$, then we have a canonical valuation $v_K : K^\times \to \mathbb{Z}$ which satisfies $v_K|_{\mathbb{Q}_p} = e_{K/\mathbb{Q}_p} v_{\mathbb{Q}_p}$.

   (a) Show that there exists a unique map $v : \overline{\mathbb{Q}}_p^\times \to \mathbb{Q}$ such that $v|_K = e_{K/\mathbb{Q}_p}^{-1} v_K$ for every finite extension $K/\mathbb{Q}_p$ such that $K \subset \overline{\mathbb{Q}}_p$.

   (b) Prove that $v : \overline{\mathbb{Q}}_p^\times \to \mathbb{Q}$ satisfies the following properties:

      i. v is surjective and $v(xy) = v(x) + v(y)$ for all $x, y \in \overline{\mathbb{Q}}_p^\times$,
      ii. $v(x + y) \geq \min(v(x), v(y))$ for all $x, y \in \mathbb{Q}_p^\times$ such that $x + y \neq 0$.

3. Let $v : \overline{\mathbb{Q}}_p^\times \to \mathbb{Q}$ be the map defined in the previous exercise, and define a metric d on $\overline{\mathbb{Q}}_p$ given by:

$$d(x, y) := \begin{cases} 2^{-v(x-y)} & \text{if } x \neq y, \\ 0 & \text{otherwise.} \end{cases}$$

   Prove that $\overline{\mathbb{Q}}_p$ is *not* complete (with respect to d).

4. Let $f(X) := 5!(1 + X + X^2/2! + X^3/3! + X^4/4! + X^5/5!) \in \mathbb{Z}[X]$.

   (a) Draw the Newton polygon of $f(X)$ over $\mathbb{Q}_p$ for each $p = 2, 3, 5$.

   (b) Prove that $f(X)$ is irreducible over $\mathbb{Q}$ and that the Galois group of the splitting field of $f(X)$ is S_5.

   [*Hint: Let E be the splitting field of $f(X)$. Compute the slopes of the relevant Newton polygons to show that $[E : \mathbb{Q}]$ is divisible by 60. Then use the discriminant of f to show that $Gal(E/\mathbb{Q}) \cong S_5$.*]